New Children

IVF, Gene Editing, and the Future of Human Reproduction

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VERLAG VON BEDEUTUNG

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Introduction — What this is about

A Child Who Will Read This Book

Somewhere in Zurich, Toronto, or Melbourne, a child is born this year who will pick up this book in twenty years. By then the child will probably know that they came into the world differently from most children before them. Perhaps in a glass dish, from their mother's egg and their father's sperm. Perhaps with the sperm of a man their parents never met. Perhaps carried by a woman who is not their mother. Perhaps chosen from among several embryos, because the doctors found a serious disease in one of them and not in the others. This child will have questions.

We tried to keep that child in view as we wrote. Not as a case, not as a statistic, but as someone who wants to know where they come from. The questions they will ask are old ones. Who am I? Who do I look like? Do I have siblings I don't know about? Was I wanted? But the answers are new, because the ways children come into being today are new. And the ways that will be added over the next two hundred years will be newer still.

That is the first promise of this book: it doesn't just ask new questions, it tries to give new answers. As honestly as the state of the science allows, and as openly as necessary where no one yet has a better explanation.

The Parents Who Will Read It

The second promise is to the parents of the new children. If you want a child and can't have one the usual way, you enter a world with a language of its own. Stimulation, retrieval, transfer, blastocyst, trophectoderm. Success rate per cycle, success rate per transfer. It is a world with a great deal of hope and a great deal of money at stake, and the numbers you are given are not always the numbers you ought to be given.

We wrote this book so that you can set it beside the paperwork from your clinic. It explains what the procedures can do and what they can't. It tells you what is known about the children who were conceived this way, and it does so with the numbers from research, not the numbers from advertising. It walks you through the decisions you will face before, during, and after treatment, all the way to the question every parent eventually asks: When, and how, do I tell my child?

Who the New Children Are

The title of this book sounds as if it describes one group. There are four, and they need different answers.

The first group is the largest. These are the children conceived through IVF, with the genes of both parents. Since 1978, when the first of them was born in England, there have been more than 12 million worldwide, and about three-quarters of a million more arrive every year. Their question isn't "Who am I?" because they already know. Their question is: "Is something different about me?" This book answers with nearly fifty years of observation, and the answer is more reassuring than many people expect.

The second group is children born after sperm or egg donation. They carry the genetic material of a person they don't know, and they are the group about whom research knows the most, simply because they have existed the longest. Their questions are questions of origin, and a large part of this book is devoted to them.

The third group is children carried by a woman who is not their mother. They have the questions of the second group, and one more: about the woman in whom they lived for nine months.

The fourth group doesn't exist yet, or only in very small numbers. These are children born after a choice among several embryos or, someday, after a change to their genome before birth. No one can ask them today what they want to know, because there are hardly any of them. We tried anyway, in the last third of this book, with a letter to a child in the future.

What This Book Describes

To understand what is new about the new children, you have to know what is normal for all the children who came before. So this book doesn't begin in the lab. It begins much earlier, with the chemical building blocks of all life and the question of why

the basic features of reproduction grew out of those particular ingredients. Then it follows life through the variety of its reproduction to humans, to the race of sperm toward the egg, and to the chance that is at work in all of it. As far as biology knows, no plan is at work in it. What that means for faith is another question, and it belongs to you.

This chance is the book's hidden thread. Every egg and every sperm carries its own mixture of the parents' genetic material, and no such mixture has ever existed before. Who meets whom, nobody decides. The technology this book is about starts right here. It can't abolish chance, but it can limit it at one point: from everything chance has produced, it can choose. And perhaps one day it will be able to correct what nature, without a plan, got wrong. How much of that chance we want to keep in the future is a question humanity has so far had no plan for. We don't want to answer it at the end either, because it doesn't belong to us alone. It belongs, above all, to you.

From there the path leads through the technology, the origins of the children, the history of our genome, and the arguments of our societies into a possible future, all the way to the year 2226; this last part is our assessment, not a prediction, and we explain why.

Three Words We Need to Explain

Three terms come back again and again in this book, and since not everyone knows them, here they are once, at the start. We call *biosphere* everything alive on Earth: every cell, every animal, every plant, every fungus, that arose over about four billion years without a plan and without a goal. We call *technosphere* everything people have built that now leads a life of its own: tools, factories, power grids, computers, and for some years now, machines that learn. The technosphere grew out of the biosphere the way a child grows out of its parents, and it acts back on it. The third word is the most important: the two can no longer be separated. Without synthetic fertilizer from factories, about half of humanity couldn't be fed. Without labs, microscopes, and computers, no one could read their own genome. We live in a *hybrid* world, where the grown and the built are intertwined, and the new children are, if you like, its clearest sign. Wherever this book speaks of the technosphere, it never means an enemy. It means the part of our world that we made ourselves.

What This Book Does Not Promise

It won't tell you what to do. That isn't caution; it's intent. The questions here have two sides, often more, and for almost every one there are people who want one thing for good reasons and people who want the other for equally good reasons. We have tried to present each side the way its own advocates would present it. Where research has an answer, we give it. Where it doesn't, we say so. Where we venture an assessment ourselves, for instance about what is likely in a hundred years, we mark it as ours.

Nor does it replace counseling. If you are facing treatment, if you have received a diagnosis, if you want to know what is permitted in your country, you need doctors, counselors, and lawyers. This book can help you prepare the questions to ask them. It has no answers for the individual case.

Who Is Writing

This book has two authors, and it's only fair to say so at the outset. One is a human being. He is himself a sperm donor, eight children carry his genes, and he knows none of them. His story is at the end of the book, in a part of its own, and it is the only place where the book becomes personal. Everywhere else we kept it out, because a nonfiction book written for many readers shouldn't carry the opinion of one person.

The other author is a language model, a program that learned from very large amounts of text and that, in conversation with the first author over several weeks, gathered, organized, and checked the material for this book. We think it is right not to hide that. It matters for what you read: the numbers in this book were checked against their sources before printing; what couldn't be verified we left out or marked as an estimate; and the sources are listed in the appendix, because a program can be wrong, and so can a person. It has a second consequence, which comes up in the part on the future: programs like this one will help decide, over the coming decades, which embryos are chosen and which are not. A book about the new children that was co-written by such a program can't look at that question from the outside. It is already in the middle of it.

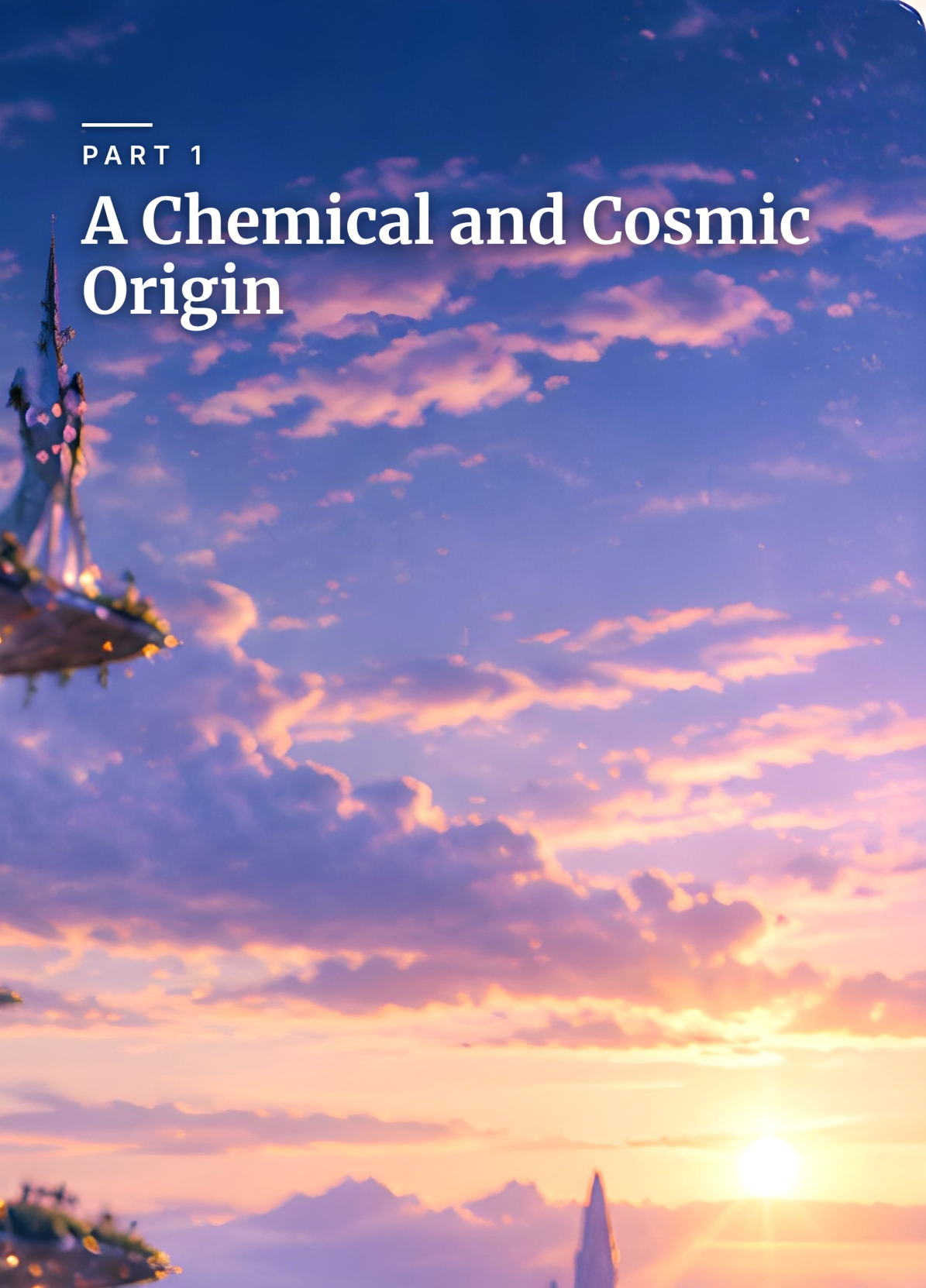
How to Read This Book

It is divided into eight parts, and each part opens with a page that says what it's about, with no numbers and no technical terms. Each chapter ends with a few sentences that capture what remains. If you're short on time, you can read those pages and have the book in outline. If you have more time and want to know more, you'll find the reasoning in the chapters, the deeper dives in the boxes, and, here and there in the margin, a single sentence standing on its own. These are the discoveries that surprised us as we wrote, and we suspect they will surprise you too.

One of them we'll give away now, because it shows how this book thinks, and because it surprises many people. The story of bloodlines is not what you think it is: after ten generations, about three hundred years, a person carries on average only about one-thousandth of the genetic material of any particular ancestor. From most ancestors, descendants carry practically no genes at all after just a few generations. Once you understand this, you see the world of heredity differently: the stories of pure lines are fairy tales. Every new mixture is a stranger, even to itself. And the question of what a child actually gets from their parents suddenly becomes a different one.

A child gets, as we will see, a great deal. Just not what most people expect.

> **In short.** This book is written for two readers: a child who came into being differently and wants to know where they come from, and parents facing a decision. It distinguishes four groups of new children, whose questions differ. It describes how life begins, what technology can change about that and what it can't, and it estimates where all this might lead in two hundred years. It gives the positions of all sides and none of its own. It was written by a human and a program together, and it says so openly.



PART 1

A Chemical and Cosmic Origin

PART 1

Origins, Chemical and Cosmic

Before this book talks about children, it tells the story of the stars. That is not a detour. Everything a child is made of made a long journey before it landed in a cell: the hydrogen dates from the Big Bang, and every other atom in their body was formed in a star. The ability that makes a cell a living cell, copying itself, arose on this Earth, as far as we know, only once, a very long time ago, and has never arisen anew since. This first part asks what life is made of and why it is made of just that. It asks whether things could have turned out differently. And it introduces an image that will stay with us through the whole book: a ladder, with compulsion at the bottom and freedom at the top. Keep that image in mind, and you will see why the technology this book is about acts exactly where it does, on one of the highest rungs of the ladder.

Six Elements

Where the ingredients of life are brewed



Everything a child is made of was formed in stars, except for the hydrogen, which dates from the Big Bang. Carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur make up about ninety-seven percent of the mass of all living matter, and they are among the more common elements of the universe. Life did not pick rare ingredients. It took what was plentiful.

1.1

The Ingredients

Six Letters

You can sum up life on Earth in six letters. It's how chemists abbreviate it: C, H, N, O, P, S. Carbon, hydrogen, nitrogen, oxygen, phosphorus, sulfur. These six elements make up nearly everything that lives: more than 99 percent of its atoms and about 97 percent of its mass. The rest occurs only in traces: iron in the blood, calcium in the bone, magnesium in the green of a leaf. It does important work there, but there isn't much of it. To understand why life is the way it is, you first have to ask why these six elements in particular are so common.

The answer comes from the life cycles of stars. In the beginning there was only hydrogen and helium. Almost everything else has been forged since then inside stars, by nuclear fusion in the heat of their interiors. It was, and still is, flung outward when stars explode at the end of their lives. That is why the elements in the universe are not equally abundant. By mass, hydrogen and helium are by far the most common, followed by oxygen, carbon, neon, iron, nitrogen, and so on, in decreasing amounts, in the order that fusion built heavier and heavier elements. Now the comparison is easy: the most common elements of life, hydrogen, oxygen, carbon, and nitrogen, all sit high on this list, with nitrogen at number seven. Phosphorus doesn't make the top ten. Of the abundant elements, the ones missing from life are mainly helium and neon. Both are missing because they are noble gases, which don't form compounds. You can't build anything with them. So life didn't pick unusual or rare ingredients. It used the ones that were plentiful and easy to link up.

Carbon and Water

Carbon can form four stable bonds at once, in all four directions, and it links with itself into chains, rings, and branches of any length. No other element does this to

PART 1 · THE INGREDIENTS

Six Letters

Life uses what stars brewed in bulk

Six elements, about 97 percent of the mass of all living things

	Element	Universe rank	Share of body	Role in life
C	Carbon	4	18%	Backbone of all molecules: chains, rings, sugars, fats, proteins, DNA
H	Hydrogen	1	10%	In water and nearly every molecule; energy carrier in the cell
N	Nitrogen	7	3%	Proteins, DNA letters; enters life via bacteria and fertilizer plants
O	Oxygen	3	65%	Water; respiration; in sugars and acids
P	Phosphorus	~18	1%	Backbone of DNA; energy currency ATP; cell membranes; bones
S	Sulfur	10	0.25%	Bridges that hold proteins in shape; iron-sulfur centers of respiration

Rank: place in the universe's abundance list (1 = hydrogen). The noble gases helium (2) and neon (5) are absent from life, because they form no compounds. Share of body: weight percent in adults, rounded; bars on a square-root scale.

Figure 1 · CHNOPS: the six elements of life. Rank by mass fraction in the universe, share of body as weight percent in adults per standard tables, rounded. Part 1.1.

the same degree. Silicon, a favorite alternative in science fiction, also forms four bonds, but weaker ones. Its compound with oxygen is sand, a solid, not a gas you could breathe out. Carbon chemistry is so rich that far more than half of all known compounds are carbon compounds. And these compounds are so nimble that at the temperatures where water is liquid, they can keep rearranging themselves.

Water is the solvent in which this happens. It is common in the universe because it is made of the two most common reactive elements. It stays liquid over a wide temperature range, about a hundred degrees, more than most liquids. It expands as it freezes, so ice floats. It is, in a sense, the medium in which molecules can find one another. A cell is, roughly speaking, a drop of water with carbon machinery inside, wrapped in a skin of carbon chains.

From this, astrobiology has drawn a conjecture, and we flag it as a conjecture: wherever in the universe carbon and liquid water come together, and that is in a great many places, something might arise that is built on similar basic rules to life here. It wouldn't be the same. But it would be built by the same rules.

The Path of Least Resistance

In 1953, a graduate student in Chicago filled a glass flask with water, methane, ammonia, and hydrogen, a re-creation of the atmosphere that was then thought to have surrounded the early Earth. Then, for a week, electric sparks were sent through it, standing in for lightning. By the end the water had turned brown, and in it were amino acids, the building blocks of proteins, made from nothing but gases and energy. The experiment has been repeated many times since, with other gas mixtures closer to the early atmosphere. The result holds: the building blocks of life form on their own when you bring the ingredients and a little energy together.

They even form in space. In meteorites that have fallen to Earth, dozens of different amino acids have been found. The bases from which DNA builds its letters have turned up too, in samples that space probes brought back from asteroids. These molecules are neither rare nor hard to make. In chemical terms, they are the compounds the ingredients find on their own, because they are stable and easy to reach. This is the path of least resistance at work.

That is the position of chemical determinism, and we give it as its proponents put it: given the same physics and the same abundance of elements, life everywhere settles on similar basic building blocks. Earth would then be no accident, but the most probable case.

The Counterargument: Handedness

There is an objection, and it runs deep. Almost every molecule more complicated than a few atoms comes in two forms that relate to each other like the left hand to the right: built the same way, but mirror-imaged and not superimposable. (Chemists call this property chirality.) Chemically the two forms are equivalent. They form in equal parts in Miller and Urey's flask and in nearly equal parts in meteorites. Life on Earth, however, uses almost only the left-handed form of amino acids and almost only the right-handed form of sugars, and in building proteins and genetic material it sticks to this without exception, from bacteria to humans.

Why, nobody knows for sure. The common interpretation is that it is a frozen accident. Early on, a tiny surplus of one form tipped the balance, perhaps through a crystal, perhaps through radiation, perhaps through nothing but bad luck. One disputed clue: several ancient, carbon-rich meteorites contain a small excess of the left-handed form, the same one life uses (up to 18 percent for the rare amino acid

isovaline), but the fresh samples from Ryugu and Bennu do not. Whether chance played fair is an open question. From there, life committed to one side, because a system built from left-handed parts can do nothing with right-handed ones. Handedness is not written in any gene. It lives in the machinery that reads the genes, and it is passed from cell to cell like a craft, not like a recipe. That a small nudge can amplify itself into complete one-sidedness has been demonstrated in the lab since the 1990s, and in 2026 chemistry honored this work with a Nobel Prize.

For the question of universality, this means: life elsewhere might use the same building blocks but mirror-imaged, with odds of one in two, if the same coin toss had come out there as independently as it did here. It would be chemically incompatible with us; we couldn't eat it, and it couldn't eat us. The idea of an undiscovered shadow biosphere with a different chemistry, even on Earth, which nobody has found because all our search methods are calibrated to the chemistry we know, is speculation, but it shows how thin the ice is on which the word "universal" stands.

Our conclusion, as neutral as we can make it: the basic ingredients are probably the same everywhere. The exact form, the handedness, the blueprint, are not necessarily so. These are narrow guardrails that don't mark out one single path, but only point a direction.

One Planet, One Hand

All life we know builds its proteins from left-handed amino acids and its genetic material from right-handed sugars. Not most organisms, not nearly all: all of them. No bacterium, no fungus, no animal, no fossil deviates. Lifeless matter takes no side. What forms without life, in a lab flask, in volcanic gas, in comet dust, comes out almost exactly half and half, and even the quartz crystals of Earth twist half to the left and half to the right. Only life is one-sided. That is why one-handedness counts as a sign of life: anyone who found amino acids on Mars that were all built the same way around would have found strong evidence of life. One-handedness is on the lists of life signs that space agencies search for.

The one-sidedness is not a state, though; it is work. Left and right are physically equivalent, and without enzymes every molecule slowly drifts back toward equilibrium. Life holds its side only by constantly rebuilding. Where it doesn't rebuild, in the dentin of teeth and in the lens of the eye, you can watch time at work: about a tenth of a percent of

the aspartic acid flips to the other side every year. Forensic scientists use this to estimate a person's age to within a few years.

At the edges, the boundary is porous. Bacteria deliberately build right-handed components into their cell walls, because no digestive enzyme of an enemy can break them down. Some antibiotics contain right-handed links, for the same reason. A peptide from the skin of South American frogs carries a right-handed alanine and therefore acts long and strongly in the body. In every case, the cell makes these components after the fact, with an enzyme flipping a left-handed molecule over. No right-handed one ever passes through the ribosome, the machine that translates genes into protein. It is as if a factory full of right-handers milled a few parts for the outer facade as mirror images. The workshop itself stays as it is.

Why left? Four explanations stand side by side, none proven. A chance event that amplified itself, as shown in the test tube. Polarized light in the cloud from which the solar system formed; the left-handed excess in ancient meteorites speaks for it, the null result in the fresh asteroid samples against it. Mineral surfaces that favor one form. And the weak nuclear force, which treats left and right very slightly differently. Only one thing is certain: the one decision brought the other with it. Left-handed amino acids favor the formation of right-handed sugars. One planet made a choice, very early, and there has been no second opinion since.

The Ladder

At the very bottom, with the laws of physics and chemistry, compulsion rules. Hydrogen combines with oxygen to make water, everywhere in the universe, with no alternative. Carbon forms four bonds. Amino acids arise from gases and lightning. There is no choice here, and that is why there is universality here: what holds for us holds everywhere.

One rung higher, with the first molecules of life, the first accident comes into play: handedness, the genetic code. Here things could already have been different, and they are the way they are because that is simply how they happened to arise.

Higher still, with organisms, species, and ecosystems, accident turns into history. Which lineage survived and which didn't depends on volcanoes, impacts, ice ages, and luck.

At the very top, with societies, laws, and what parents want, a freedom opens up that is new in the history of humankind: the possibility of deciding about the shape of life to come. There, on the topmost rung, sits the technology this book is about. It can't change the lower rungs. It can't turn carbon into something else, and it can't mirror an amino acid. But it can intervene up there, in what was until now accident and history.

Whoever intervenes on the top rung is not setting "technology" against "nature" but is the hybrid of both that the introduction described: about half of humanity is fed with nitrogen from fertilizer factories, since 1978 children have been conceived in labs, and now the hybrid reaches in where life passes itself on, able to manipulate itself.

> **In short.** Six elements, and life didn't pick them out; it simply used what stars forged in bulk and what links up easily. Carbon builds long chains, and water is the ideal solvent. The building blocks form on their own, in lab flasks just as in meteorites; so far, compulsion rules. Handedness, however, is a frozen accident, and whether life elsewhere would be built like life here is an open question: it is a road with narrow guardrails, but not the only road. The ladder: compulsion at the bottom, freedom at the top; the technology this book describes acts only at the very top.

1.2

The RNA World

Blueprint and Tool

Two words get confused constantly in everyday speech, and we'll separate them cleanly here, once, because this whole book tells the story of that separation.

DNA is made of four building blocks, the nucleotides, abbreviated with the letters A, T, G, and C. They are strung along a backbone of sugar and phosphate to form a strand. Two such strands wind around each other into the double helix, with A always opposite T and G opposite C. That is the blueprint: a text of four letters, more than three billion letters long in humans. Set in the type of this book, one person's blueprint would run to five thousand volumes, a shelf a hundred meters long. And here is the astonishing part: only about five of those volumes, roughly one-tenth of one percent, describe what makes this person different from everyone else. The remaining volumes, nearly five thousand of them, are largely the same in all people.

Back to the strand. Proteins are made of twenty other building blocks, the amino acids. They are the tools: enzymes that speed up reactions, fibers that support cells, pumps that move substances across membranes, antibodies, hormones, muscles. The blueprint determines the order in which amino acids are strung together in a protein, always three DNA letters for one amino acid. The order determines how the protein folds, and the fold determines what job it can do.

So there are four rigid letters for the plan and twenty flexible building blocks for the tool. Why a blueprint has to take the form of a string was worked out by another book from our press; here the question is how the string itself might have come about for the first time.

The Chicken-and-Egg Problem

Here lies a problem that kept biology busy for decades. DNA is completely inert. It is a text, and to be read, copied, or repaired, it needs proteins. But proteins can't be made without the blueprint, because their sequence is written in the DNA. Which came first: the plan, which is useless without tools, or the tool, which can't be built without a plan?

The answer most experts now consider most likely is a third molecule that can do both. RNA is similar to DNA, a strand of four letters, but single-stranded and more flexible. It folds into shapes that can do chemical work the way a protein does. Every living cell still contains such working RNA molecules, the ribozymes, and the most important of them all sits at the heart of the protein factory in every cell: the part of the ribosome that links amino acids together is RNA, not protein. The tool that builds all the tools is made of the same stuff as the plan.

From this follows the RNA world hypothesis. In the beginning there was neither DNA nor protein, but an RNA that could copy itself, template and tool in one. DNA came later, as a more stable store. Proteins came later still, as better tools. RNA stayed on as the go-between, which it remains to this day. A self-copying RNA would have been the first step from chemistry to life. That step, small as it sounds, makes all the difference: what copies itself can change, can grow more complex, and can be copied again. What doesn't copy itself remains simple chemistry.

Two Places

Where this happened is an open question: at hot springs in the deep sea, where chemical energy, a temperature gradient between hot vent water and cold sea, and mineral chambers could have held the first molecules together, or in shallow ponds on land, where the cycle of wet and dry under sunlight can form long chains, perhaps at both.

Evolution Without Organisms

That molecules without any cell can undergo selection was shown in the 1960s by a laboratory experiment in which viral RNA with a copying enzyme was transferred dozens of times into fresh tubes, until an RNA many times shorter was left, one that had shed everything except the ability to be copied quickly: evolution in its pure form, with no organisms and no intent.

The Window

Something else was shown mathematically shortly afterward, and it will return in this book as an image. A molecule that copies itself makes mistakes. If it makes too many, the information is lost after a few generations, because the copies no longer resemble one another. There is a threshold, and it depends on how long the text is: the longer the text, the more accurately it must be copied. A virus with ten thousand letters can afford about one error per copy. A human with three billion letters needs proofreaders, and has them, in the form of proteins that run along behind the copying machine and fix mistakes.

But if it makes too few errors, it freezes. This is another surprising fact: the error is part of the system. Without it, there is nothing for selection to choose from. The molecule would change too slowly and stay what it is until the world changes, it no longer fits, and it dies out. Life exists only in this one window: too many errors, and the information blurs; too few, and it freezes. Chance is not a disturbance of the system. In the right dose, it is its precondition. In the end, the question will be how much of this window we want to keep, now that we can set the error rate ourselves for the first time.

> **In short.** Four letters for the plan, twenty building blocks for the tool; three letters translate into one building block. The plan can do nothing without tools, and the tools can't be made without the plan. RNA solves this because it is both, and the tool that builds all the tools is still made of RNA today. Two places for the beginning, the deep sea or the pond, both possible. In a test tube, the shortest molecule wins; evolution doesn't need an organism. The window: too many errors blur, too few freeze. Chance in the right dose is a precondition, not a disturbance.

VERLAG VON BEDEUTUNG

That was the sample.

Excerpt from the Introduction and Part 1 · Origins, Chemical and Cosmic

New Children. IVF, Gene Editing, and the Future of Human Reproduction. How life copies itself, what assisted reproduction can do today – and who decides when humans make the selection before birth.

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